AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Administration

FIN301-F
Financial & Securities Markets
Version B

Final Examination
Spring 2025-2026

2 – June – 2026

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Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 Hours**

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF"** mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examination
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

Solve All questions

Question 1: (30 marks)

a) A $1,000 par bond with an annual coupon has only one year until maturity. Its current yield is 7.621%, and its yield to maturity is 12%. What is the price of the bond?

b) Suppose Microsoft, Inc. was trading at $27.29 per share. At that time, it pays an annual dividend of $0.32 per share, and analysts have set a 1-year target price around $33.30 per share. What is the expected return on this stock?

Question 2: (40 marks)

a) Compute the required monthly payment on a $80,000 30-year, fixed-rate mortgage with a nominal interest rate of 5.80%. How much of the payment goes toward principal and interest during the first year? Provide an amortization schedule till the 12th month.

b) Suppose you buy a call option on a $100,000 Treasury bond futures contract with an exercise price of 105. If the price of the Treasury bond is 115 at expiration, is the option at the money, in the money, or out of the money? Determine the premium if the profit equals $8,000.

Question 3: (30 marks)

What is a mortgage loan? Explain four types of mortgage loans.

End of Questions

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Formulas

Tax-free rate= taxable interest rate × (1 – marginal tax rate)

$$P_0 = \frac{D_0(1+g)}{k_e - g}$$

$$P = \frac{r(PV)}{1-(1+r)^{-n}}$$

$P = Payment$
$PV = Present\ Value$
$r = rate\ per\ period$
$n = number\ of\ periods$

$$\frac{\Delta P}{P} = -\text{Duration} \times \frac{\Delta i}{1+i}, \quad \text{or} \quad \text{Duration} = -\frac{1+i}{\Delta i} \times \frac{\Delta P}{P}$$

$$\boldsymbol{i_c = C / P}$$

Equivalent tax-free rate (ETFR)= (1-MTR)

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